

1.5 Introduction to HTML & HTML Features:

What is HTML?

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

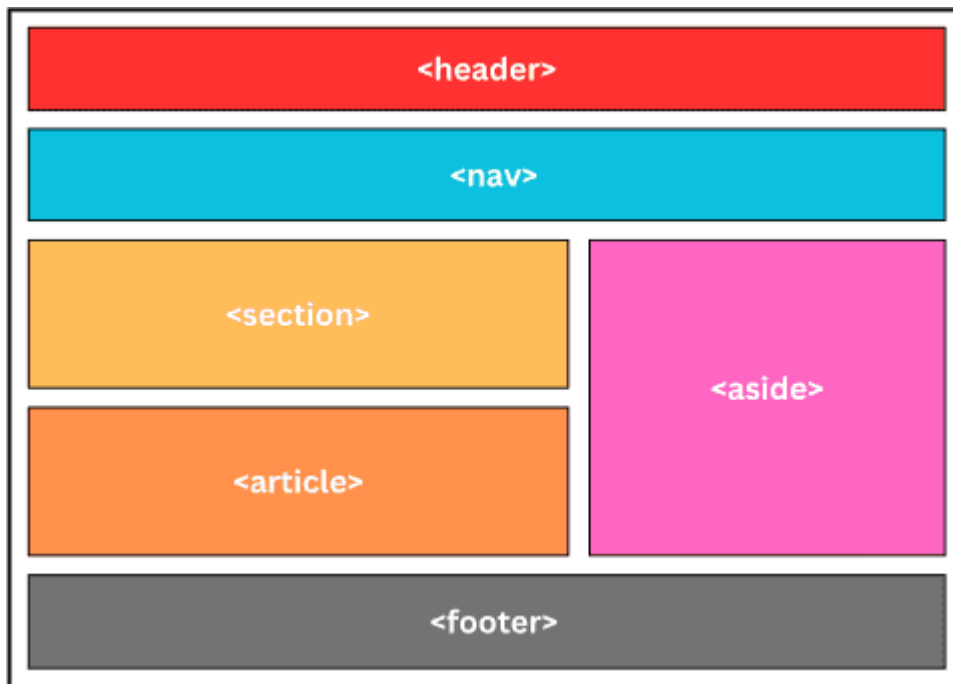
HTML 5 Features

HTML5 was introduced having some new features following are

- Error tolerance
- Save as html or .htm
- Pass data in “.....” or ‘.....’
- Name Casing <HTML> or <html>
- Semantic Elements
- Audio and Video Support
- Canvas Elements
- Geolocation API
- Local Storage
- Responsive Images
- Web Workers
- Drag and Drop API
- Form Enhancements
- Web Sockets
- Micro Data

1.6 HTML Page Structure:

An HTML page is structured using a hierarchical arrangement of elements, providing a logical organization for the content. The fundamental structure includes:



`<!DOCTYPE html>` Declaration:

This declaration, placed at the very beginning of an HTML document, informs the browser that the document is an HTML5 document. It is not an HTML tag but a directive.

`<html>` Element:

This is the root element of an HTML page, encapsulating all other HTML elements. It signifies the beginning and end of the HTML document.

`<head>` Element:

Located within the `<html>` element, the `<head>` contains metadata about the HTML document. This information is not directly displayed on the web page but is crucial for browser rendering, search engine optimization, and other functionalities. Common elements within the `<head>` include:

`<title>`: Defines the title of the document, which appears in the browser tab or window title bar.

`<meta>`: Provides various metadata, such as character set (charset), viewport settings, and descriptions for search engines.

`<link>`: Links external resources like stylesheets (.css files) or favicons.

`<style>`: Embeds internal CSS styles directly within the HTML document.

`<script>`: Embeds or links external JavaScript code for interactive functionality.

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Basic Structure Page</title>

</head>

<body>

  <h1>Welcome to HTML</h1>

</body>

</html>
```

<body> Element:

Also located within the <html> element, the <body> contains all the visible content of the web page. This is where the actual content that users interact with is placed. Examples of elements found within the <body> include:

Headings (<h1> to <h6>): Define titles and subtitles for sections of content.

Paragraphs (<p>): Enclose blocks of text.

Links (<a>): Create hyperlinks to other web pages or resources.

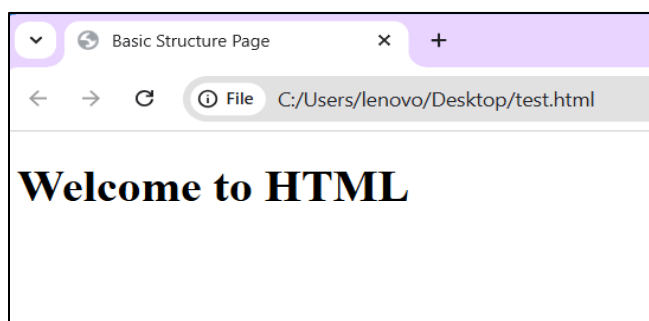
Images (): Embed images into the page.

Lists (, ,): Create unordered or ordered lists of items.

Divisions (<div>): Generic container for grouping other elements for styling or scripting purposes. Provide more meaningful structure to the content, improving accessibility and SEO.

This nested structure, starting from the <!DOCTYPE html> and encompassing the <html>, <head>, and <body> elements, forms the fundamental framework of every HTML page.

Output:



1.7 Elements in HTML(Semantic and Non Semantic)

A tag is a keyword that defines how a web browser should format and display a web page. HTML tags are the basic building blocks of any website. In html, codes(normal text) enclosed in brackets written in usually paired.

See below example:

`<TITLE>My Web Page</TITLE>`

Note: - HTML is not case sensitive. `<TITLE>` = `<title>` =

`<TitLE>` In HTML,

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Basic Sturcture Page</title>

  </head>

  <body>

    <header>Website Header</header>

    <nav>Main Navigation</nav>

    <main>

      <article>

        <h2>Blog Post Title</h2>

        <p>This is the content of the blog post...</p>

      </article>

      <section>

        <h2>About Us</h2>

        <p>We are a company that builds great products.</p>

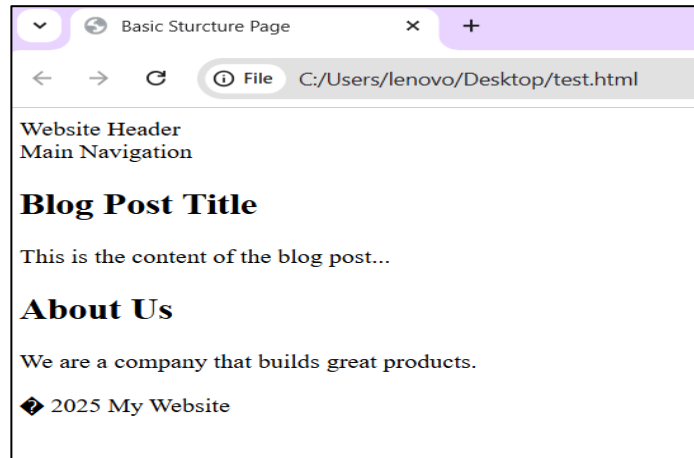
      </section>

    </main>

    <footer>© 2025 My Website</footer>

  </body>
```

Output:



Semantic tags define the meaning of the content they contain, while non-semantic tags only hold content and do not indicate its role on the page.

Semantic tags are both human- and machine-readable, and they can help users and machines understand the context and importance of the content.

Non-semantic tags, on the other hand, are used for styling purposes or as containers for other elements.

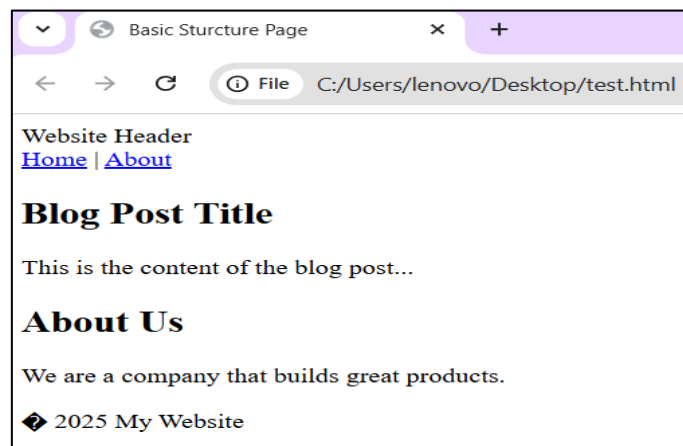
For Example: the `<div>` tag does not suggest the content it will carry, however using the `<p>` tag suggests it can be used to hold paragraph information.

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Basic Sturcture Page</title>
  </head>
  <body>
    <div id="header">Website Header</div>
    <div id="menu"><a href="#">Home</a> | <a href="#">About</a></div>
    <div id="content">
      <div class="post">
        <h2>Blog Post Title</h2>
        <p>This is the content of the blog post...</p>
      </div>
      <div class="info">
        <h2>About Us</h2>
        <p>We are a company that builds great products.</p>
      </div>
    </div>
    <div id="footer">© 2025 My Website</div>
  </body>

```

Output:



There are many reasons why you should write Semantic tags instead of normal HTML tags, to leverage SEO, accessibility, and browser compatibility.

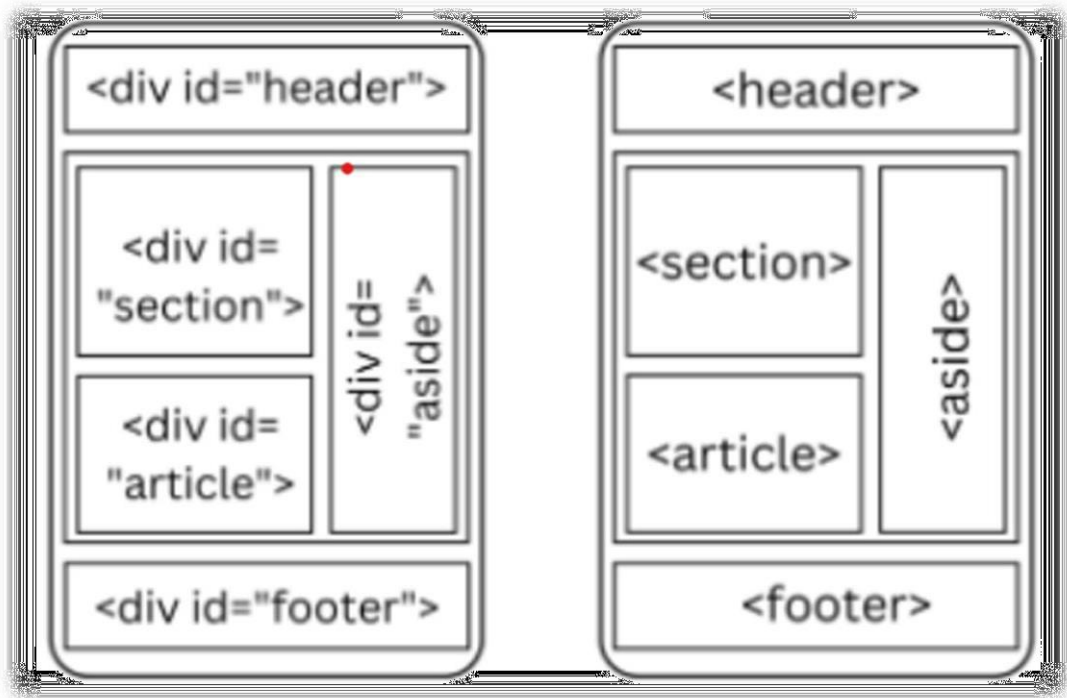


Fig: Non-Semantic and Semantic Tags